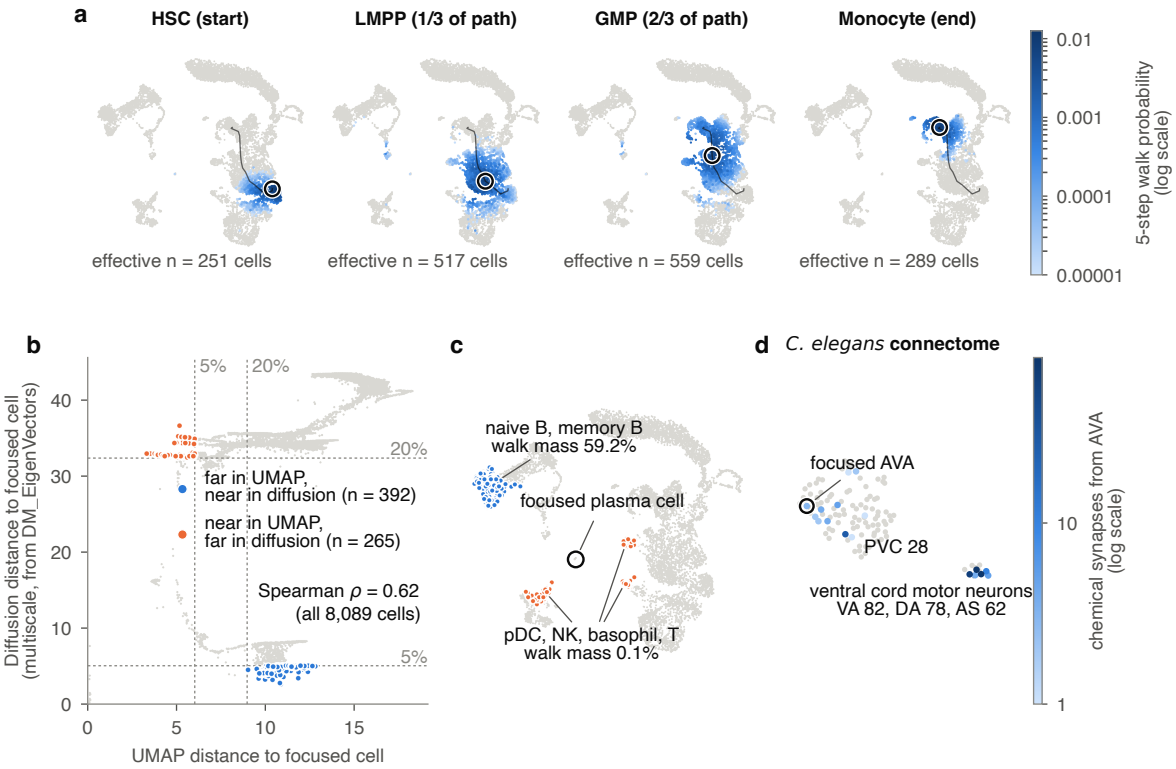

a
HSC (start)
LMPP (1/3 of path)
GMP (2/3 of path)
Monocyte (end)
effective n = 251 cells
effective n = 517 cells
effective n = 559 cells
effective n = 289 cells
0.01
0.001
0.0001
0.00001
5-step walk probability (log scale)
b
5%
20%
20%
5%
far in UMAP, near in diffusion (n = 392)
near in UMAP, far in diffusion (n = 265)
Spearman ρ = 0.62 (all 8,089 cells)
Diffusion distance to focused cell (multiscale, from DM_EigenVectors)
UMAP distance to focused cell
0
5
10
15
20
30
40
c
naive B, memory B walk mass 59.2%
focused plasma cell
pDC, NK, basophil, T walk mass 0.1%
d
C. elegans connectome
focused AVA
PVC 28
ventral cord motor neurons VA 82, DA 78, AS 62
chemical synapses from AVA (log scale)
1
10